# EXPIREMENT-01

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| Ex.No. 1 | **Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.** |
|---|---|
| Date: | |

**AIM:** - To Study of various Network commands used in Linux and Windows

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| **1** | `arp -a` | `arp -a` | Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network. | `arp -a` | `Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic` |
| **2** | `hostname` | `hostname` | Displays the hostname (computer name) of the local system. | `hostname` | `DESKTOP-PC01` |
| **3** | `ipconfig /all` | `ipconfig /all` | Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address. | `ipconfig /all` | `IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes` |
| **4** | `nbtstat -a` | `nbtstat -a <hostname>` | Displays NetBIOS name table of a remote computer for | `nbtstat -a DESKTOP-PC01` | `NetBIOS Remote Machine Name Table Name Type Status DESKTOP-PC01 <00> UNIQUE Registered` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | troubleshooting NetBIOS over TCP/IP. | | |
| **5** | `netstat` | `netstat` | Displays active TCP/IP connections, listening ports, routing tables, and interface statistics. | `netstat` | `Proto Local Address Foreign Address State TCP 192.168.1.10:49762 142.250.190.78:443 ESTABLISHED` |
| **6** | `netstat -r` | `netstat -r` | Displays the IP routing table of the local computer. | `netstat -r` | `IPv4 Route Table Network Destination Netmask Gateway Interface` |
| **7** | `nslookup` | `nslookup <domain>` | Queries the DNS server to obtain the IP address or DNS records of a domain. | `nslookup www.google.com` | `Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68` |
| **8** | `pathping` | `pathping <hostname/IP>` | Combines the features of `ping` and `tracert` to trace the route and measure packet loss at each hop. | `pathping www.google.com` | `Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0%` |
| **9** | `ping` | `ping <hostname/IP>` | Tests connectivity between the local computer and another host using ICMP Echo Request and Reply messages. | `ping www.facebook.com` | `Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35: bytes=32 time=25ms TTL=54` |
| **10** | `route` | `route print` | Displays or modifies the IP routing table. It is commonly used to view or | `route print` | `IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | configure static routes. | | |

## Some important Linux networking commands

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| 1 | ip | `ip <OPTIONS> <OBJECT> <COMMAND>` | Displays and manages IP addresses, routing tables, and network interfaces. | `ip address show` | `2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0` |
| 2 | ip address add | `ip address add <IP>/<Prefix> dev <Interface>` | Assigns an IP address to a network interface. | `ip address add 192.168.1.254 /24 dev eth0` | `Assigned 192.168.1.254/24 to eth0 successfully.` |
| 3 | ip address del | `ip address del <IP>/<Prefix> dev <Interface>` | Removes an IP address from a network interface. | `ip address del 192.168.1.254 /24 dev eth0` | `Deleted 192.168.1.254/24 from eth0.` |
| 4 | ip link set up | `ip link set <Interface> up` | Brings the network interface online. | `ip link set eth0 up` | `Interface eth0 is UP.` |
| 5 | ip link set down | `ip link set <Interface> down` | Brings the network interface offline. | `ip link set eth0 down` | `Interface eth0 is DOWN.` |
| 6 | ip link set promisc on | `ip link set <Interface> promisc on` | Enables promiscuous mode on the interface. | `ip link set eth0 promisc on` | `Promiscuous mode enabled on eth0.` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| 7 | ip route add default | ip route add default via <Gateway> dev <Interface> | Adds the default gateway for all outgoing traffic. | ip route add default via 192.168.1.254 dev eth0 | Default route added successfully. |
| 8 | ip route add | ip route add <Network> via <Gateway> | Adds a static route through a gateway. | ip route add 192.168.1.0/24 via 192.168.1.254 | Route added successfully. |
| 9 | ip route add (Direct Route) | ip route add <Network> dev <Interface> | Adds a direct route through a network interface. | ip route add 192.168.1.0/24 dev eth0 | Route added via eth0. |
| 10 | ip route delete | ip route delete <Network> via <Gateway> | Deletes a static route. | ip route delete 192.168.1.0/24 via 192.168.1.254 | Route deleted successfully. |
| 11 | ip route get | ip route get <Destination IP> | Displays the route used to reach a destination. | ip route get 10.10.1.4 | 10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100 |
| 12 | ifconfig | ifconfig | Displays and configures network interface parameters (deprecated in modern Linux). | ifconfig | eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0 |
| 13 | mtr | mtr <hostname/IP> | Combines Ping and Traceroute to | mtr google.com | HOST: localhost Loss% Snt Last Avg Best Wrst StDev |

| | | | | | |
|---|---|---|---|---|---|
| | | | analyze network paths. | | |
| 14 | mtr -g | `mtr -g <hostname>` | Displays numeric IP addresses instead of hostnames. | `mtr -g google.com` | `192.168.1.1 → 142.250.190.78` |
| 15 | mtr -b | `mtr -b <hostname>` | Displays both hostnames and IP addresses. | `mtr -b google.com` | `router.local (192.168.1.1) google.com (142.250.190.78)` |

| | | | | | |
|---|---|---|---|---|---|
| 16 | mtr -c | `mtr -c <count> <hostname>` | Sends a specified number of probes. | `mtr -c 10 google.com` | `10 probes completed successfully.` |
| 17 | tcpdump -D | `tcpdump -D` | Lists all available network interfaces. | `tcpdump -D` | `1.eth0 2.lo 3.wlan0` |
| 18 | tcpdump | `tcpdump -i <Interface>` | Captures packets on a network interface. | `tcpdump -i eth0` | `Listening on eth0...` |
| 19 | tcpdump -c | `tcpdump -i <Interface> -c <count>` | Captures a specified number of packets. | `tcpdump -i eth0 -c 10` | `10 packets captured.` |
| 20 | tcpdump host | `tcpdump -i <Interface> host <IP>` | Captures traffic to and from a specific host. | `tcpdump -i eth0 host 8.8.8.8` | `Captured packets to/from 8.8.8.8.` |
| 21 | tcpdump src host | `tcpdump -i <Interface> src host <IP>` | Captures packets originating from a specific host. | `tcpdump -i eth0 src host 8.8.8.8` | `Source packets from 8.8.8.8 displayed.` |

| | | | | | |
|---|---|---|---|---|---|
| 22 | tcpdump dst host | `tcpdump -i <Interface> dst host <IP>` | Captures packets destined for a specific host. | `tcpdump -i eth0 dst host 8.8.8.8` | `Destination packets to 8.8.8.8 displayed.` |
| 23 | tcpdump net | `tcpdump -i <Interface> net <Network>/<Prefix>` | Captures traffic from a specified network. | `tcpdump -i eth0 net 10.1.0.0/24` | `Packets from network 10.1.0.0/24 displayed.` |
| 24 | tcpdump port | `tcpdump -i <Interface> port <Port>` | Captures traffic on a specific port. | `tcpdump -i eth0 port 53` | `DNS packets captured.` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| 1 | ip | `ip <OPTIONS> <OBJECT> <COMMAND>` | Displays and manages IP addresses, routing tables, and network interfaces. | `ip address show` | `2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0` |
| 2 | ip address add | `ip address add <IP>/<Prefix> dev <Interface>` | Assigns an IP address to a network interface. | `ip address add 192.168.1.254/24 dev eth0` | `Assigned 192.168.1.254/24 to eth0 successfully.` |
| 3 | ip address del | `ip address del <IP>/<Prefix> dev <Interface>` | Removes an IP address from a network interface. | `ip address del 192.168.1.254/24 dev eth0` | `Deleted 192.168.1.254/24 from eth0.` |
| 4 | ip link set up | `ip link set <Interface> up` | Brings the network interface online. | `ip link set eth0 up` | `Interface eth0 is UP.` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| 5 | ip link set down | ip link set <Interface> down | Brings the network interface offline. | ip link set eth0 down | Interface eth0 is DOWN. |
| 6 | ip link set promisc on | ip link set <Interface> promisc on | Enables promiscuous mode on the interface. | ip link set eth0 promisc on | Promiscuous mode enabled on eth0. |
| 7 | ip route add default | ip route add default via <Gateway> dev <Interface> | Adds the default gateway for all outgoing traffic. | ip route add default via 192.168.1.254 dev eth0 | Default route added successfully. |
| 8 | ip route add | ip route add <Network> via <Gateway> | Adds a static route through a gateway. | ip route add 192.168.1.0/24 via 192.168.1.254 | Route added successfully. |
| 9 | ip route add (Direct Route) | ip route add <Network> dev <Interface> | Adds a direct route through a network interface. | ip route add 192.168.1.0/24 dev eth0 | Route added via eth0. |
| 10 | ip route delete | ip route delete <Network> via <Gateway> | Deletes a static route. | ip route delete 192.168.1.0/24 via 192.168.1.254 | Route deleted successfully. |
| 11 | ip route get | ip route get <Destination IP> | Displays the route used to reach a destination. | ip route get 10.10.1.4 | 10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100 |
| 12 | ifconfig | ifconfig | Displays and configures network interface parameters | ifconfig | eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0 |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | (deprecated in modern Linux). | | |
| 13 | mtr | `mtr <hostname/IP>` | Combines Ping and Traceroute to analyze network paths. | `mtr google.com` | `HOST: localhost Loss% Snt Last Avg Best Wrst StDev` |
| 14 | mtr -g | `mtr -g <hostname>` | Displays numeric IP addresses instead of hostnames. | `mtr -g google.com` | `192.168.1.1 → 142.250.190.78` |
| 15 | mtr -b | `mtr -b <hostname>` | Displays both hostnames and IP addresses. | `mtr -b google.com` | `router.local (192.168.1.1) google.com (142.250.190.78)` |
| 16 | mtr -c | `mtr -c <count> <hostname>` | Sends a specified number of probes. | `mtr -c 10 google.com` | `10 probes completed successfully.` |
| 17 | tcpdump -D | `tcpdump -D` | Lists all available network interfaces. | `tcpdump -D` | `1.eth0 2.lo 3.wlan0` |
| 18 | tcpdump | `tcpdump -i <Interface>` | Captures packets on a network interface. | `tcpdump -i eth0` | `Listening on eth0...` |
| 19 | tcpdump -c | `tcpdump -i <Interface> -c <count>` | Captures a specified number of packets. | `tcpdump -i eth0 -c 10` | `10 packets captured.` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| 20 | tcpdump host | `tcpdump -i <Interface> host <IP>` | Captures traffic to and from a specific host. | `tcpdump -i eth0 host 8.8.8.8` | `Captured packets to/from 8.8.8.8.` |
| 21 | tcpdump src host | `tcpdump -i <Interface> src host <IP>` | Captures packets originating from a specific host. | `tcpdump -i eth0 src host 8.8.8.8` | `Source packets from 8.8.8.8 displayed.` |
| 22 | tcpdump dst host | `tcpdump -i <Interface> dst host <IP>` | Captures packets destined for a specific host. | `tcpdump -i eth0 dst host 8.8.8.8` | `Destination packets to 8.8.8.8 displayed.` |
| 23 | tcpdump net | `tcpdump -i <Interface> net <Network>/<Prefix>` | Captures traffic from a specified network. | `tcpdump -i eth0 net 10.1.0.0/24` | `Packets from network 10.1.0.0/24 displayed.` |
| 24 | tcpdump port | `tcpdump -i <Interface> port <Port>` | Captures traffic on a specific port. | `tcpdump -i eth0 port 53` | `DNS packets captured.` |
| 25 | tcpdump host and port | `tcpdump -i <Interface> host <IP> and port <Port>` | Captures traffic for a specific host and port. | `tcpdump -i eth0 host 8.8.8.8 and port 53` | `DNS traffic with host 8.8.8.8 displayed.` |
| 26 | tcpdump port not | `tcpdump -i <Interface> port not <Port1> and not <Port2>` | Captures all traffic except the specified ports. | `tcpdump -i eth0 port not 53 and not 25` | `Traffic excluding ports 53 and 25 captured.` |
| 27 | ping | `ping <hostname/IP>` | Tests network connectivity using ICMP Echo Requests | `ping google.com` | `64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms` |

| S.No | Command | Synopsis | Description | Sample Input | Sample Output |
|---|---|---|---|---|---|
| | | | and Echo Replies. | | |
| 28 | ping -c | `ping -c <count> <hostname>` | Sends a specified number of ICMP Echo Requests. | `ping -c 10 google.com` | `10 packets transmitted, 10 received, 0% packet loss.` |

**Configuring an Ethernet connection by using nmcli:**

| Step No. | Task | Command | Expected Result |
|---|---|---|---|
| **1** | List all Network Manager connection profiles | `nmcli connection show` | Displays all available network connection profiles with their **Name, UUID, Type, and Device**. |
| **2** | Create a new Ethernet connection profile<br><br>It is specifically an **`nmcli` connection management command**.<br><br>• **`nmcli`** → NetworkManager Command Line Interface<br>• **`connection`** → Manages network connection profiles.<br>• **`add`** → Creates a new connection profile. | `nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet` | Creates a new Ethernet connection profile named **Internal-LAN**. |
| **3** | Rename an existing connection profile (Optional) | `nmcli connection modify "Wired connection 1" connection.id Internal-LAN` | Renames the existing connection profile to **Internal-LAN**. |

| Step No. | Task | Command | Expected Result |
|---|---|---|---|
| **4** | Display the current settings of the connection profile | `nmcli connection show "Internal-LAN"` | Displays interface name, autoconnect status, IPv4 and IPv6 configuration details. |
| **5** | Configure IPv4 using DHCP | `nmcli connection modify "Internal-LAN" ipv4.method auto` | Configures the interface to obtain the IPv4 address automatically using DHCP. |
| **6** | Configure a static IPv4 address | `nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com` | Assigns a static IPv4 address, gateway, DNS server, and DNS search domain. |
| **7** | Configure IPv6 using SLAAC | `nmcli connection modify "Internal-LAN" ipv6.method auto` | Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC). |
| **8** | Configure a static IPv6 address | `nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffee ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com` | Assigns a static IPv6 address, gateway, DNS server, and search domain. |
| **9** | Activate the connection profile | `nmcli connection up Internal-LAN` | Activates the **Internal-LAN** network connection profile. |
| **10** | Verify the IP address configuration | `ip address show enp1s0` | Displays the assigned IPv4 and IPv6 addresses of the **enp1s0** network interface. |
| **11** | Verify the IPv4 default gateway | `ip route show default` | Displays the configured IPv4 default gateway. |
| **12** | Verify the IPv6 default gateway | `ip -6 route show default` | Displays the configured IPv6 default gateway. |

| Step No. | Task | Command | Expected Result |
|---|---|---|---|
| **13** | Verify the DNS configuration | `cat /etc/resolv.conf` | Displays the configured DNS servers and DNS search domain. |
| **14** | Test network connectivity | `ping <host-name-or-IP-address>` | Sends ICMP Echo Requests to verify connectivity with the specified host. |

**Sample Verification Outputs**

| Verification Command | Sample Output |
|---|---|
| ip address show enp1s0 | inet 192.0.2.1/24<br>inet6 2001:db8:1::fffe/64 |
| ip route show default | default via 192.0.2.254 dev enp1s0 proto static |
| ip -6 route show default | default via 2001:db8:1::ffee dev enp1s0 proto static |
| cat /etc/resolv.conf | search example.com<br>nameserver 192.0.2.200<br>nameserver 2001:db8:1::ffbb |
| ping google.com | 64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms |

**RESULT:**

**Thus,**

**Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.**

```
File  Edit  View  Search  Terminal  Help
student@fedora:~$ mtr -g google.com
mtr: invalid option -- 'g'

Usage:
 mtr [options] hostname

 -F, --filename FILE        read hostname(s) from a file
 -4                         use IPv4 only
 -6                         use IPv6 only
 -u, --udp                  use UDP instead of ICMP echo
 -T, --tcp                  use TCP instead of ICMP echo
 -I, --interface NAME       use named network interface
 -a, --address ADDRESS      bind the outgoing socket to ADDRESS
 -f, --first-ttl NUMBER     set what TTL to start
 -m, --max-ttl NUMBER       maximum number of hops
 -U, --max-unknown NUMBER   maximum unknown host
 -P, --port PORT            target port number for TCP, SCTP, or UDP
 -L, --localport LOCALPORT  source port number for UDP
 -s, --psize PACKETSIZE     set the packet size used for probing
 -B, --bitpattern NUMBER    set bit pattern to use in payload
 -i, --interval SECONDS     ICMP echo request interval
 -G, --gracetime SECONDS    number of seconds to wait for responses
 -Q, --tos NUMBER           type of service field in IP header
 -e, --mpls                 display information from ICMP extensions
```

```
student@fedora:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolve
d(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but onl
y
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
```

```
File  Edit  View  Search  Terminal  Help
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<
global>
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 138  bytes 29665 (28.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 168  bytes 17540 (17.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 2130 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 2130 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
File  Edit  View  Search  Terminal  Help
                        My traceroute  [v0.95]
fedora (10.0.2.15) -> google.com (142.250.182.78)    2026-07-25T09:09:18+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                  Packets               Pings
 Host                          Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. lcmaaa-ax-in-f14.1e100.net  0.0%     3    8.1   8.8   8.1   9.7   0.8
```

```
Command Prompt
C:\Users\DELL>nbtstat -a HDC0124105

Ethernet 2:
Node IpAddress: [192.168.56.1] Scope Id: []

    Host not found.

Ethernet:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

WiFi:
Node IpAddress: [172.16.75.45] Scope Id: []

    Host not found.

Local Area Connection* 9:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

Local Area Connection* 10:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []

    Host not found.

C:\Users\DELL>
```

```
student@fedora:~$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolve
d(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but onl
y
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
```

```
student@fedora:~1$ nmcli connection show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  86ead386-e6ab-359e-9d82-71491148899f  ethernet  enp0s3
lo                  d3bc006d-3b51-4d5e-ba62-b097234a30ab  loopback  lo
student@fedora:~$
```

```
C:\Users\DELL>pathping HDC0124105

Tracing route to HDC0124105 [172.16.75.45]
over a maximum of 30 hops:
  0  HDC0124105 [172.16.75.45]
  1  HDC0124105 [172.16.75.45]

Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                          HDC0124105 [172.16.75.45]
                                 0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  HDC0124105 [172.16.75.45]

Trace complete.

C:\Users\DELL>
```

```
File  Edit  View  Search  Terminal  Help
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<
global>
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 138  bytes 29665 (28.9 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 168  bytes 17540 (17.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 2130 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 2130 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
student@fedora:~$ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp0s
3
10.0.2.0        0.0.0.0         255.255.255.0   U         0 0          0 enp0s
3
student@fedora:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.77.110
Name:   google.com
Address: 2404:6800:4007:802::200e

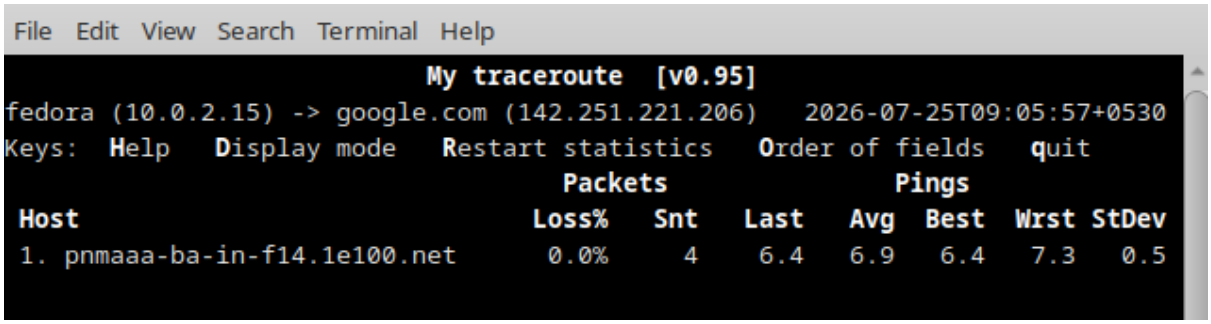
student@fedora:~$
```

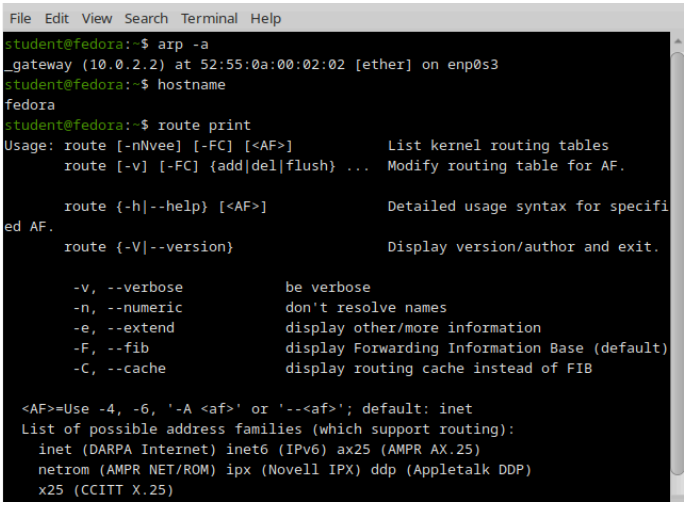
```
File  Edit  View  Search  Terminal  Help
student@fedora:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

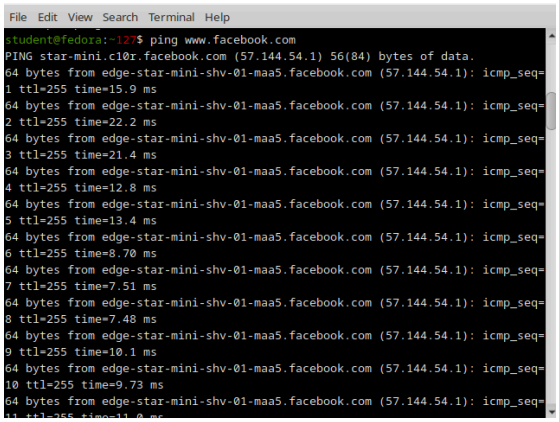
udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHE
D
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     19174    /run/systemd/journa
l/stdout
unix  3      [ ]         STREAM     CONNECTED     17819    /run/systemd/journa
l/stdout
unix  3      [ ]         STREAM     CONNECTED     23930
unix  3      [ ]         STREAM     CONNECTED     22896
unix  3      [ ]         STREAM     CONNECTED     22825    /run/systemd/journa
l/stdout
unix  3      [ ]         STREAM     CONNECTED     22567    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     19354    /run/dbus/system_bu
s_socket
unix  3      [ ]         STREAM     CONNECTED     22624    /run/dbus/system_bu
s_socket
unix  3      [ ]         STREAM     CONNECTED     22817    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22591    /run/dbus/system_bu
s_socket
```

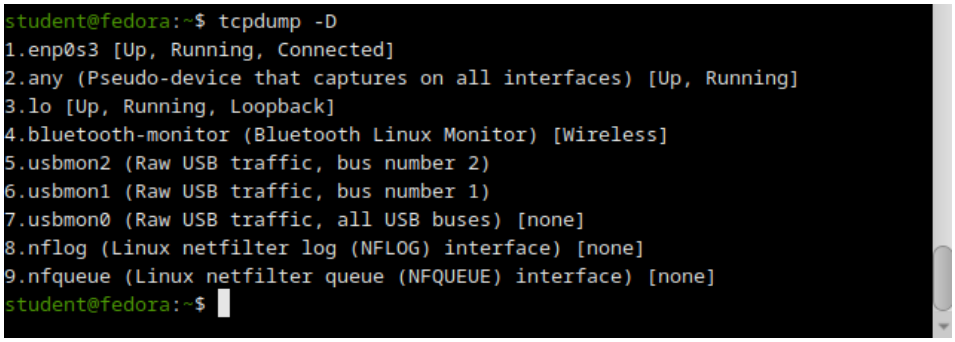
```
student@fedora:~1$ nmcli connection show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  86ead386-e6ab-359e-9d82-71491148899f  ethernet  enp0s3
lo                  d3bc006d-3b51-4d5e-ba62-b097234a30ab  loopback  lo
student@fedora:~$
```

```
student@fedora:~127$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defa
ult qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83835sec preferred_lft 83835sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefi
xroute
       valid_lft 86135sec preferred_lft 14135sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
student@fedora:~$
```


File Edit View Search Terminal Help
My traceroute [v0.95]
fedora (10.0.2.15) -> google.com (142.251.221.206) 2026-07-25T09:05:57+0530
Keys: Help Display mode Restart statistics Order of fields quit
Packets Pings
Host Loss% Snt Last Avg Best Wrst StDev
1. pnmaaa-ba-in-f14.1e100.net 0.0% 4 6.4 6.9 6.4 7.3 0.5


File Edit View Search Terminal Help
student@fedora:~$ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
student@fedora:~$ hostname
fedora
student@fedora:~$ route print
Usage: route [-nNvee] [-FC] [<AF>] List kernel routing tables
route [-v] [-FC] {add|del|flush} ... Modify routing table for AF.
route {-h|--help} [<AF>] Detailed usage syntax for specifi
ed AF.
route {-V|--version} Display version/author and exit.
-v, --verbose be verbose
-n, --numeric don't resolve names
-e, --extend display other/more information
-F, --fib display Forwarding Information Base (default)
-C, --cache display routing cache instead of FIB
<AF>=Use -4, -6, '-A <af>' or '--<af>'; default: inet
List of possible address families (which support routing):
inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
x25 (CCITT X.25)


File Edit View Search Terminal Help
student@fedora:~127$ ping www.facebook.com
PING star-mini.c10r.facebook.com (57.144.54.1) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=1 ttl=255 time=15.9 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=2 ttl=255 time=22.2 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=3 ttl=255 time=21.4 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=4 ttl=255 time=12.8 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=5 ttl=255 time=13.4 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=6 ttl=255 time=8.70 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=7 ttl=255 time=7.51 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=8 ttl=255 time=7.48 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=9 ttl=255 time=10.1 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=10 ttl=255 time=9.73 ms
64 bytes from edge-star-mini-shv-01-maa5.facebook.com (57.144.54.1): icmp_seq=


student@fedora:~$ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
9.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
student@fedora:~$